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**Statement of Purpose**

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**I want to design next-generation spatial user interfaces and intelligent tools.** This idea originated in my growth during my middle school years when the mobile internet was starting to boom, and I was deeply inspired by design innovations of electronic products such as the iPhone. This interest led me to a multidisciplinary education at ShanghaiTech University, where I studied CS, design, and humanities, and further refined my skills and taste through research at UIUC and Harvard. I find designing human-centered interfaces critical for effective technology application, and exploring design challenges through various research methods is fascinating to me. For these reasons, I am determined to pursue a Ph.D. in HCC at Georgia Tech.

Sparks of Artificial General Intelligence (AGI) and emerging spatial computing platforms provide a vast design space for Human-Computer Interaction, and three challenges in it excite me the most. First, spatial computing platforms break the boundaries of traditional screen-based interfaces, calling for customized information input and display methods to maximize the high interaction freedom in 3D space. Second, as Large Language Models (LLMs) begin to show AGI capabilities, the boundaries between human and AI abilities have become increasingly blurred, thus human-AI collaboration requires a more precise division of roles to support human values such as creativity. Lastly, as rapid technology advancement imposes higher demands on people's literacy and adaptability to new things, we need to design user interfaces for technologically underprivileged groups—transforming technology into new opportunities for personal growth rather than barriers. Over the past few years, I have conducted research on several projects spanning these themes, primarily focused on human-centered user interface design. For my Ph.D. studies, I want to take a broader perspective, while still addressing the design challenges around the above three themes.

**Exploring intuitive interaction design for spatial computing.** Recognizing that human gaze reveals spatial attention during the interaction, I explored using eyes as an input source for spatial computing headsets and led a research project on VR gaze interaction. Existing gaze interaction methods mainly use gaze direction for target pointing, but struggle with Midas Touch problem for target selection—distinguishing between 'looking' and 'choosing' intents. The key problem is that gaze direction does not naturally contain reliable information to differentiate these two intentions; therefore an additional input signal for selection is needed. After analyzing the design space of eye input, we discovered that visual depth calculated from binocular vision is an intriguing dimension. For instance, we can look through a window at a distant landscape or focus closer to observe the dust on the window—we can intentionally change how far we see to selectively observe different objects. Inspired by this, we placed a 'virtual window' in front of users, so they can choose an object by bringing their focus closer and displaying the information on the window. This idea led to FocusFlow, a hands-free gaze selection method using visual depth shift for VR headsets. We also designed multiple user studies to evaluate the usability and learnability of this novel eye input technique, and found that providing immediate and clear visual feedback can greatly help users understand and master this interaction process. This intuitive and efficient gaze-depth interaction method has demonstrated application value in both general and professional scenarios and was showcased as a demo at UIST 2023, with the full version currently under review. [project page]

**Supporting human creativity in future human-AI interaction.** The advent of LLMs has provided a powerful technological foundation for human-AI interaction. During the summer of 2023, I explored the collaboration patterns between designers and LLMs at the Harvard Visual Computing Group. We aim to leverage the LLM's fuzzy matching capability to translate UI

designers' design principles into environment-customized visualization parameters, thereby enhancing UI adaptability to context. This process revealed a dual ambiguity in describing environmental conditions and human needs, posing challenges in communication between LLMs and designers. While vague design principles ease designers' burden, they may lead to unstable model decisions. In contrast, specific descriptions heavily rely on preset conditions provided by the designers, diminishing the LLMs' inferential value. To address this, we proposed a JSON-structured declarative natural language grammar for conveying design constraints to LLMs, and explored ways to define and balance the ambiguity in human-AI natural language communication. From this research, I realized that we need to rethink human-AI values and offer a more human-centered mindset for AI applications. [project page]

I also approached HCI artifact values from the creativity perspective, understanding how tools can aid the creative process. Joining Professor Sarah Sterman's newly formed lab focusing on creativity research, I engaged in qualitative research on the roles of student documentation in Physical Computing education. We first analyzed course syllabi across various majors and educational levels using the open coding method, developing an understanding of the motivations, tools, content, and learning goals for student-created documentation. Further, we conducted semi-structured interviews with course instructors from various majors to dig into their values on documentation and challenges faced in education practice. This hands-on research experience opened me to qualitative research methodologies, enhancing my ability to derive insights from textual data and learn directly from our users. [project page]

**Making technology accessible to more people.** Growing up in a small city in China with limited resources, I benefited from the transformative power of HCI artifacts symbolized by digital devices mobile internet in the 2010s. While appreciating the opportunities technology provided, I also recognized its challenges. During an educational outreach experience in Southwest China, I realized the complex impact of technology. Despite the widespread Internet access enabled by government initiatives, the information divide persists in underdeveloped areas. Children are often dangerously exposed to low-quality sensory-charged online content, and local teachers struggle to integrate multimedia technology effectively in education. These experiences highlighted the limitations of current HCI in marginalized communities, fueling my motivation to develop more inclusive technological solutions.

**During my doctoral studies, I aim to integrate the above three topics to explore user interface and intelligent tool design in spatial computing scenarios.** Specifically, I hope to research telepresence technology in mixed reality, enabling users physically in different spaces or times to communicate and collaborate as if they were in the same place. This involves fundamental research questions about how to reconstruct the spatial and temporal context of immersive communication, and interact with objects in virtual environment. Moreover, spatial computing transforms human-AI collaboration space from 2D screens to 3D environments. How to create more context-aware AI systems, and enabling more end-users to intuitively create customized and flexible virtual tools in mixed reality excites me greatly. Georgia Tech is an ideal research environment for me, where its extensive HCI community can offer a solid foundation for my academic journey. In particular, the research of Professor Yalong Yang in display and interaction techniques between 2D and 3D user interfaces, along with Professor Thad Starner's work in mixed reality interactive systems and multi-sensory interfaces greatly inspire my spatial computing research. Additionally, the research conducted by Professors HyunJoo Oh and Cindy Xiong in the fields of human-centered design, creativity, and cognitive processes also offers wonderful perspectives for my artifact research. I hope to design intelligent user interfaces to address challenging problems in spatial computing within the School of Interactive Computing's interdisciplinary research context, combining different technologies and research methods, and thus contributing new possibilities to the field of Human-Computer Interaction.